

JORDAN ZUGE

MECHANICAL ENGINEER - NAVAL HVAC & SYSTEMS DESIGN

CONTACT

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 Marinette, WI

EDUCATION

University of Wisconsin Stout

B.S. Mechanical Engineering
Menomonie, WI

SKILLS & CERTIFICATIONS

Shipboard & Mechanical Systems:

- HVAC Functional Design & Pressure Loss Optimization
- SSDG Exhaust & Gas Turbine Intake System Design
- Mechanical Component & Controls Integration
- Heat Transfer & System Dynamics

Design & Modeling Software:

- AutoCAD
- ShipConstructor
- Navisworks
- SolidWorks & SolidWorks PDM
- Autodesk Inventor

Engineering Documentation:

- Test Procedure Development
- Ship Check & Foundation Verification
- Technical Documentation & Drawings
- Design Reviews

Programming & Analysis:

- MATLAB
- Python
- EES

Project & Communication Tools:

- Microsoft Office Suite
- Teams, Zoom, Visio, MS Project

PROFILE

Mechanical Engineer with experience in naval HVAC system design, shipboard integration, ship checks, foundation verification, and test procedure development. Skilled in 3D modeling, routing, fabrication drawings, and multidisciplinary coordination to deliver MIL-STD compliant shipboard systems.

PROFESSIONAL EXPERIENCE

HVAC Functional/Design Engineer

TMMG @ Fincantieri Marinette Marine

May 2025 - Present

- Designed and routed complex shipboard HVAC ducting, reducing pressure losses, improving airflow efficiency, and improving crew habitability.
- Produced fabrication and installation drawings that enabled accurate first-pass installations and minimized waterfront rework.
- Performed HVAC pressure loss analyses to optimize system balancing, energy efficiency, and equipment life.
- Resolved HVAC-to-structure, HVAC-to-piping, and HVAC-to-electrical clashes in 3D models, which resulted in early conflict elimination that benefited production schedule protection.
- Developed 3D CAD models for SSDG exhaust and gas turbine intake systems, resulting in fabrication-ready routing packages that supported on-time equipment installation.
- Developed and authored shipboard HVAC system test procedures to support equipment verification, commissioning, and MIL-STD compliant acceptance testing.
- Performed ship checks to verify equipment foundations and installed arrangements, resolving field discrepancies between production, structural, and HVAC design to minimize rework.
- Collaborated with production engineering during design reviews, reducing field change requests through proactive issue resolution.

Mechanical Engineering Internship

Rice Lake Weighing Systems

May 2024 - August 2024

- Worked directly with CNC machining teams, resulting in design-for-manufacturability improvements that reduced cycle times.
- Designed and manufactured a production jig, improving workflow efficiency and daily output consistency.
- Created production-ready parts, assemblies, and drawings for a series of pipe lever scales, resulting in fabrication packages that supported both internal manufacturing and customer deliveries.
- Led the integration of a newly acquired company's parts library into EPDM, resulting in standardized part numbering and data control that improved enterprise inventory management and documentation integrity.

ORGANIZATIONS

Kappa Lambda Beta Co-ed Fraternity

Membership: January 2022 - December 2024

Positions Held:

- Served as Fraternity President, coordinating academic, service, and fundraising initiatives that increased member participation and strengthened organizational stability.
- Directed fundraising programs that increased financial support for Mothers Against Drunk Driving and chapter operations, enabling expanded community outreach and event programming, including biweekly highway cleanups.